DM 2

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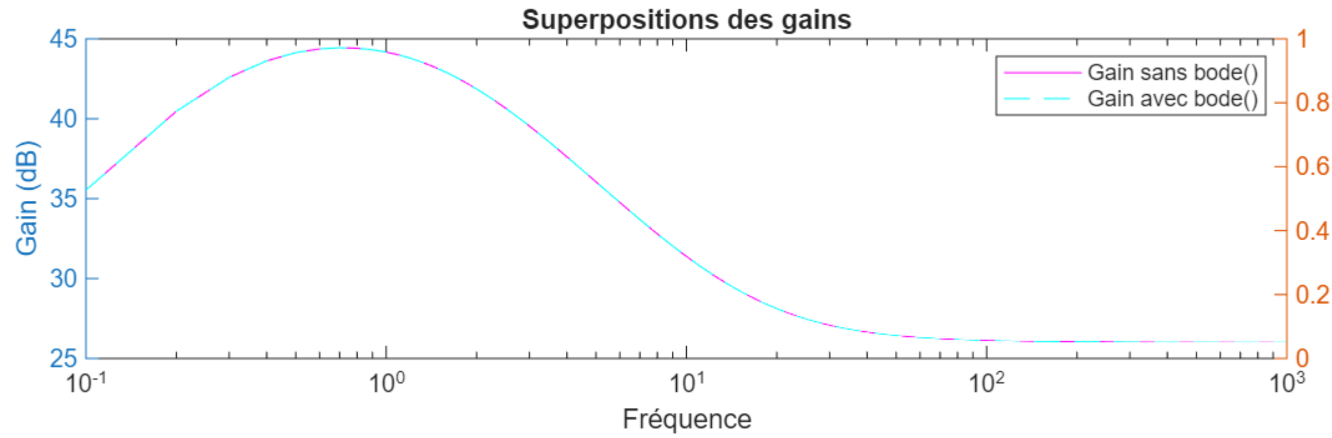
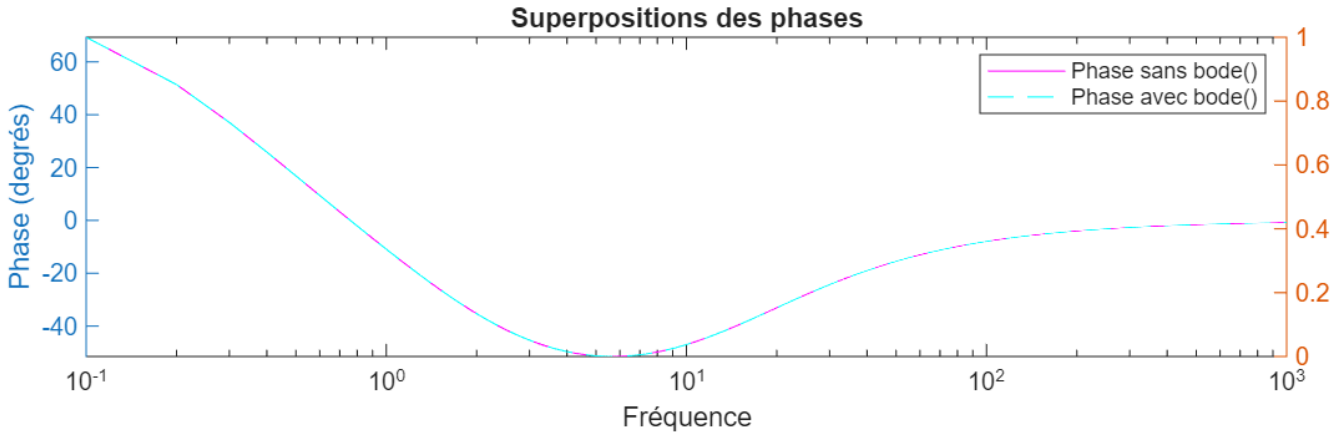
Exercice 1





Code correspondant :

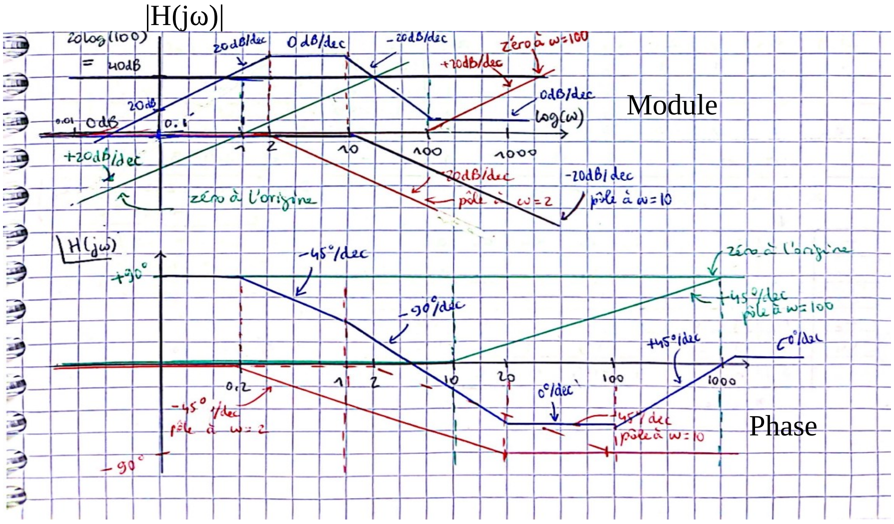
```
% sans bode()
f = 0.1:0.1:1000;
s = 1j*2*pi*f;
H1 = (20 * s .* (s + 100)) ./ ((s + 2) .* (s + 10));

gain1 = 20 * log10(abs(H1)); %en dB
phase1 = angle(H1) * (180/pi); %en degrés

% avec bode()
H = tf([20, 2000], [1, 12, 20]);
[gain2, phase2, wout] = bode(H, 2 * pi * f);
```

Version manuscrite :

|H(jω)|



Exercice 2

1) $X(S) = \dfrac{10(S+1)}{S^2+4S+3} = \dfrac{10(S+1)}{(S+1)(S+3)} = \dfrac{A}{(S+1)} + \dfrac{B}{(S+3)}$

Trouvons A : $\dfrac{10(S+1)\cancel{(S+1)}}{\cancel{(S+1)}(S+3)} = \dfrac{A\cancel{(S+1)}}{\cancel{(S+1)}} + \dfrac{B(S+1)}{(S+3)}$

Prenons $S=-1$:

$$\frac{10\overbrace{(-1+1)}^{0}}{(-1+3)} = A + \frac{B\overbrace{(-1+1)}^{0}}{(-1+3)}$$

$0 = A + 0$, donc, $\boxed{A=0}$

Trouvons B :

$$\frac{10(S+1)\cancel{(S+3)}}{(S+1)\cancel{(S+3)}} = \frac{A(S+3)}{(S+1)} + \frac{B\cancel{(S+3)}}{\cancel{(S+3)}}$$

Prenons $S=-3$:

$$\frac{10(-3+1)}{(-3+1)} = \frac{A\overbrace{(-3+3)}^{0}}{(S+1)} + B$$

$\dfrac{-20}{-2} = 10 = 0 + B$, donc, $\boxed{B = 10}$

Ainsi, $\dfrac{10(S+1)}{S^2+4S+3} = \dfrac{10}{S+3} = X(S) \xrightarrow{\mathcal{L}^{-1}} \boxed{x(t) = 10\, e^{-3t} u(t)}$

2) $X(S) = \dfrac{10(S+1)\,e^{-2S}}{S^2+4S+3} = \dfrac{10(S+1)\,e^{-2S}}{(S+1)(S+3)} = \left(\dfrac{A}{(S+1)} + \dfrac{B}{(S+3)}\right)e^{-2S}$

Trouvons A : $\dfrac{10(S+1)\quad\cancel{(S+1)}}{\cancel{(S+1)}(S+3)} = \dfrac{A\cancel{(S+1)}}{\cancel{(S+1)}} + \dfrac{B(S+1)}{(S+3)}$

Prenons $S=-1$: $\dfrac{10(-1+1)}{(-1+3)} = A + \dfrac{B\overbrace{(-1+1)}^{0}}{(S+3)}$

$0 = A + 0$, donc $\boxed{A=0}$

Trouvons B :

$$\frac{10(S+1)\quad\cancel{(S+3)}}{(S+1)\cancel{(S+3)}} = \frac{A(S+3)}{(S+1)} + B\frac{\cancel{(S+3)}}{\cancel{(S+3)}}$$

Prenons $S=-3$: $\dfrac{10(-3+1)}{(-3+1)} = \dfrac{A(-3+3)}{(-3+1)} + B$

$\dfrac{-20}{-2} = 10 = 0 + B$, donc $\boxed{B = 10}$

Ainsi, $X(S) = \dfrac{10(S+1)e^{-2S}}{S^2+4S+3} \xrightarrow{\mathcal{L}^{-1}} \boxed{x(t) = 10 \quad e^{-3(t-2)} \quad u(t-2)}$

3) $X(S) = \dfrac{20}{S(S^2+10S+16)} = \dfrac{20}{S(S+2)(S+8)} = \dfrac{A}{S} + \dfrac{B}{(S+2)} + \dfrac{C}{(S+8)}$

Trouvons A : $\dfrac{20\cancel{S}}{\cancel{S}(S+2)(S+8)} = \dfrac{A \times \cancel{S}}{\cancel{S}} + \dfrac{B \times S}{(S+2)} + \dfrac{C \times S}{(S+8)}$

Prenons $S = 0$: $\dfrac{20}{2 \times 8} = A + 0 + 0$

$\dfrac{20}{16} = \dfrac{5}{4} = A$, donc $\boxed{A = \dfrac{5}{4}}$

Trouvons B : $\dfrac{20\cancel{(S+2)}}{S\cancel{(S+2)}(S+8)} = \dfrac{A(S+2)}{S} + \dfrac{B\cancel{(S+2)}}{\cancel{(S+2)}} + \dfrac{C(S+2)}{(S+8)}$

Prenons $S = -2$: $\dfrac{20}{-2(-2+8)} = \dfrac{A\overbrace{(-2+2)}^{0}}{-2} + B + \dfrac{C\overbrace{(-2+2)}^{0}}{(-2+8)}$

$\dfrac{20}{-12} = 0 + B + 0$, donc $\boxed{B = -\dfrac{5}{3}}$

Trouvons C : $\dfrac{20\cancel{(S+8)}}{S(S+2)\cancel{(S+8)}} = \dfrac{A(S+8)}{S} + \dfrac{B(S+8)}{(S+2)} + \dfrac{C\cancel{(S+8)}}{\cancel{(S+8)}}$

Prenons $S = -8$: $\dfrac{20}{-8(-8+2)} = \dfrac{A(-8+8)}{-8} + \dfrac{B(-8+8)}{(-8+2)} + C$

$\dfrac{20}{48} = \dfrac{5}{12} = 0 + 0 + C$, donc $\boxed{C = \dfrac{5}{12}}$

Ainsi, $X(S) = \dfrac{20}{S(S^2+10S+16)} = \dfrac{5}{4S} - \dfrac{5}{3(S+2)} + \dfrac{5}{12(S+8)}$

$\mathcal{L}^{-1}$

$\boxed{x(t) = \dfrac{5}{4}u(t) - \dfrac{5}{3}e^{-2t}u(t) + \dfrac{5}{12}e^{-8t}u(t)}$